

ÍG‘İmK<Ä]åQw

~yvît %œ±hc

N N*WÚNŽ Node.js + Express + SQLite v„W(˘ÍG‘İmK<Ä|û~ß

N 0 ~yvîi,•đ

g,~yvîf/N N*—bT [fu v„W(˘ÍG‘İmK<Ä]åQwÿ u(b7SiNå••ÇVP{TSU<ÍÊNI•bé~ge<ÄO0•ê]ñv„,ÍIGI4^s0 |û~ße/c \ [f

1.1 ~yvîvîh

- cĐO>N yÍg •£0 šØeHv„,ÍIG‘İ<ÄO0e1_
- Ç PK [ùb XžR [fN`R`R>TÆ•£Ts`
- <°_Uu(b7[fN`•h•ùÿ ^.R©N†%œ±•êŽ«•Ůke
- e/c •~g,RŸÉÿÿ]éVú...„_1wå<Æp¹

1.2 vîh u(b7

- W(h![fu Ÿ \ [f0 R N-0 šØN-ÿ
- ^ g biQE,ñ<ÍÍG‘İv„,[fN`€
- %œ•|û~ß<ÄO0•êŽ«<ÍIGI4^sv„,u(b7

NŒ0 b€g/gŦg„

2.1 b€g/h

0 RMzİb€g/0

- HTML5 + CSS3 + JavaScript (ES6+)
- Bootstrap 5 UI hFgŦ
- TÍ^”_ <¾<ıÿ e/c yûR`zi

0 T zİb€g/0

- Node.js •Đ^Leö
- Express.js Web hFgŦ
- SQLite epcn^“ (••Ç Sequelize ORM)
- JWT (JSON Web Token) Ž«Ný<¤<Á
- bcrypt [ÆEx R [Æ

0 _ SŦ]åQw0

- VS Code Năx ••Vh
- Git rHg,c§R6
- Postman API mK<Ů

2.2 |û~ßgŦg„Vp

|û~ß‘Çu(RMT zİR y»gŦg„ÿ

u(b7mO%ŒVh !•!' HTTP/REST API !•!' Express.js g RjVh !•!' SQLite epcn^“

2.3 ~yvîvî_U~Óg,,

vocabulary-test/

- % % % app.js # ^”u(QeSãe‡Nö
- % % % vocabulary_test.html # RMziN;~u—b
- % % % embedded_word_data.js # <í^“epcn
- % % % config/ # ‘M•ne‡Nö
- % % % % database.js # epcn^“M•n
- % % % models/ # epcnj!W<
- % % % % User.js # u(b7j!W<
- % % % % Word.js # SU<Íj!W<
- % % % % TestRecord.js # mK<Õ<°_Uj!W<
- % % % routes/ # •ü1e‡Nö
- % % % % user.js # u(b7vøQsc¥Sã
- % % % % word.js # SU<ÍvøQsc¥Sã
- % % % % test.js # mK<ÕvøQsc¥Sã
- % % % % admin.js # {it TXc¥Sã
- % % % % pk.js # PK[ùb c¥Sã
- % % % middleware/ # N-•ôNö
- % % % % auth.js # <¤<ÁN-•ôNö
- % % % utils/ #]åQwQýep
- % % % % jwt.js # JWT]åQw
- % % % data/ # <í^“epcne‡Nö

N 0 h8_ÃRŸ€ýj!WW

3.1 u(b7<¤<Á|û~ß

[žs°[Œetv,,u(b7lèQŒ0 v{ _U0 v{QúRŸ€ýŸ

- u(b7lèQŒÿ e/c u(b7T TŒ[Æx lèQŒÿ [Æx O•u(bcrypt R [Æ[XP”
- u(b7v{ _Uÿ šŒ<ÁQí<ÁT •ÔVÞ JWT Token
- Token {it Ÿ RMzi[XP”W(localStorageÿ <÷lBeö•êR”d: ^&
- •êR”v{ _Uÿ ~u—bR7e°eö•êR”`bY v{ \_Ur¶`
- gC–PS:R Ÿ fn• u(b7TŒ{it TX%ŒÖ,r

3.2 <ÍG‘İmK<Ä

h8_ÄmK<ÄRŸ€ýŸ S T+NåN ryp¹Ÿ

- N ~š<í^“Ÿ \ [f(505<Í)0 R N-(1110<Í)0 šØN-(1866<Í)
- fz€ýQú”Ÿ h9cnu(b7_SRMI4^sR”” Œ et”vî–¾4^İ
- Y • ” _b_ Ÿ kİ”6N*• ~y(1kcxn+4^rbp+N <¤<Æ)
- [žeö~ß<iŸ kcxnep0 • <iep0 kcxns‡
- ~Óg_gaNöŸ [Œb 50” / •Þ~í{T• 4” / }/<i{T• 7”
- <ÍG‘İ<ÄO0Ÿ WÚNŽkcxnep<i{—O0{—<ÍG‘İ

3.3 AI [ùb j!_

N AI [ùbK[žeo[ùb ÿ

- N hc–¾^|ÿ {€SU0 N-{I0 Vđ–¾
- AI j!bßÿ h9cn–¾^|THs°N T {T~• ^|TÆkcxns‡
- b¢{Tg:R6ÿ QH{T[ù€ _—R
- –Peö{T~ÿ kÎ~30yÒP <|eö
- [žëöKÔR f>y:

3.4 u(b7S9‘M PK

w N°[ùb RŸ€ýÿ

- b?•ôS9‘Mÿ u(b7SîR ^úb R Qeb?•ô
- [žëöT key ~vîTÆeö•ôT kef>y:
- b¢{T_—R g:R6
- €Ü• R\$[šÿ {T~epTÆkcxns‡~üt <ÄO0

3.5 • ~g,|û~ß

- ~e6–ÆEN Y N`ÿ
- •êR`e6–Æÿ mK<Ä•Çz N-{T• v„SU<Í•êR`R Qe• ~g,
- • ~Y N`ÿ e/c • <ÍXkzzj!\_ Y N`
- yû–dg:R6ÿ Y N`kcxnT •êR`NÎ• ~g,yû–d
- g,W0[XP`ÿ O•u(localStorage c NES

3.6 N*N°N- _Ã

u(b7RŸ€ý€ZT ÿ

- N*N°Oá`oÿ u(b7T 0 %ò,r0 {l~§
- mK<Ä~ß<|ÿ mK<Äk!ep0 g šØ<ÍG`î0 ^sWGkcxns‡
- S†Sò<°_Uÿ gâw S†k!mK<Ä~Ógœ
- l4^sSØS ÿ <ÍG`îSØS •<R¿Vp`h
- • ~g,{it ÿ gâw TÆY N`• ~

3.7 {it TXT Sđ

|û~ß{it RŸ€ýÿ

- u(b7{it ÿ gâw u(b7R ^h0 y•u(/T/u(•&S÷
- <Í^“{it ÿ XžR e9gâSU<Íepcn
- mK<Ö<°_Uÿ gâw b@g u(b7v„mK<Ä<°_U
- epcn~ß<|ÿ |û~ßO•u(epcn~ß<|

VÛ0 epcn^“<¾<|

4.1 u(b7^h (Users)

[Wkµ<ôf ÿ

- id: N;•ÿ •êXž
- username: u(b7T ÿ U/N

- password: [Æx (bcryptR [Æ)
- role: %oÖ,r(user/admin)
- createdAt: R ^úeö•ô
- updatedAt: fêe°eö•ô

4.2 SU<Í^h (Words)

[Wkμ<ôf ÿ

- id: N;•.ÿ •êXž
- word: SU<Í
- meaning: N-e±'ÊNI
- level: -¾^!~\$R+(primary/middle/high)
- createdAt: R ^úeö•ô

4.3 mK<Ö<°_U^h (TestRecords)

[Wkμ<ôf ÿ

- id: N;•.ÿ •êXž
- userId: Y •.ÿ Qs€Tu(b7
- score: _—R
- vocabulary: O0{—<ÍIG'í
- accuracy: kcxns±
- questionCount: {T~~ep
- createdAt: R ^úeö•ô

N”0 API c¥Sãe±hc

5.1 u(b7c¥Sã

POST /api/users/register - u(b7lèQÆ

POST /api/users/login - u(b7v{ _U

GET /api/users/profile - f·SÖu(b7Oá`o(— <¤<Á)

5.2 SU<Íc¥Sã

GET /api/words - f·SÖSU<ÍR ^h

POST /api/words - mûR SU<Í({it TX)

PUT /api/words/:id - fêe°SU<Í({it TX)

DELETE /api/words/:id - R -dSU<Í({it TX)

5.3 mK<Öc¥Sã

POST /api/tests/submit - cÐNªmK<Ö~Ógœ

GET /api/tests/history - f·SÖ\$†Sò<°_U(— <¤<Á)

5.4 {it TXc¥Sã

GET /api/admin/users - u(b7R ^h

PUT /api/admin/users/:id - f0e°u(b7r¶`

GET /api/admin/words - <Í^“{jt

GET /api/admin/tests - mK<Ö<°_U

5.5 PK c¥Să

POST /api/pk/create - R ^úPKb?•ô

POST /api/pk/join - R QePKb?•ô

GET /api/pk/status/:id - f·SÖPKr¶`

POST /api/pk/answer - cĐN▯{ThH

Qm0 [%0Qh<¾<i

6.1 [Æx [%0Qh

- O•u(bcrypt {—lÖR [Æ[XP`[Æx
- R vĐY t ý -2kb_ity^he;Qû
- v{ UY1•%k!ep-PR6

6.2 Ž«Ný<▯<Á

- JWT Token •Çg eö•ôÿ 7Y)
- Token [XP`W(localStorage
- kĭk!<÷lB•êR`d:^& Authorization Y4
- Token •Çg •êR`•ó•lv{ U~u

6.3 gC-Pc§R6

- {jt TXc¥Să— %0• admin %0Ö,r
- u(b7c¥Să— %0•g eH Token
- -2kb•ŠgC<¿•îQvNÖu(b7epcn

N 0 •è•r<ôf

7.1 s¯Xf%0•lB

- Node.js >= 14.x
- npm >= 6.x
- s°NămO%0ÈVh(Chrome0 Firefox0 Edge0 Safari)

7.2 [%0^Åkeš▯

1. QK-t~yvî

```
git clone https://gitee.com/wang-yuchangabc0/word-test.git
cd word-test
```

2. [%0^ÅO••V

```
npm install
```

3. T/R`g Rj

```
npm start
```

4. <¿•ŕ^"u(

m0%ÈVhbS_ http://localhost:3001

7.3 žØ«•&S÷

- {jt TXÿ admin / admin123
- fn• u(b7ÿ — •ê~LlèQÆ

Qk0 bi\URŸ€ý^ú<®

8.1 wíg O S

- mûR •®NöleQÆšÆ<Á
- [žs°u(b7Y4PĪN O
- XžR fôY <ĪG~šR+TÆN N ~†Wß
- O S yûR~ziOSšÆ

8.2 N-g %oÄR

- WebSocket [žeö• Oá(u(NžPK[ùb)
- c^~Liœ|û~ß
- kĭeåbSSa/[fN`<ijR RŸ€ý
- Y <íŠ e/c

8.3 •g a?fo

- Al fz€ýc`fP[fN`•i_,,
- <í—óg <ûTÆT,Q™RŸ€ý
- y>S:N'R`RŸ€ý
- yûR`zi App
- epcnR g•Nê~hvØ

N]0 ^8%oÁ•ĩ~

Q1: Y,OU'í•n{jt TX[Æx ÿ

A: SiNå• •Ç Node.js • g,vôc¥Oîe9epcn^"N-v,,[Æx [Wkµÿ O•u(bcrypt 'Íe°R [Æe°[Æx 0

Q2: Y,OUXžR e°v,,SU<íR0<í^“ÿ

A: v{ _U{jt TXT Sđÿ W(" <í^" {jt ~"u—bmûR e°SU<íÿ b • •Ç API c¥Săby'ĩ[üQe0

Q3: e/c TêN›mO%ÈVhÿ

A: e/c Chrome0 Firefox0 Edge0 Safari v,,g e°N\$N*rHg,0

Q4: epcn[XW(Tê'ÿ

A: b@g epcn[XP`W(SQLite epcn^"e‡Nö database.sqlite N-ÿ Sivôc¥Y Ný<ăe‡Nö0